An exact retuning of a 20,000-cylinder tetrahedron cover

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Abstract

We give an explicit rational parameter choice in the compatible-secant construction of van de Kamp, itself based on OpenAI's angular cylinder construction. For the closed regular tetrahedron of edge length two, 20,000 infinite cylinders with compact nondegenerate triangular perpendicular bases have total base area, normalized by the minimum projection area, strictly less than 0.4999213. This improves the checked public 20,000-cylinder benchmark 0.499921585997001774. The parameter changes from $e = 1/10$ to $e = 13/125$; the geometric family and its padding rule are retained. We reproduce the closed-set coverage proof on its full domain $0 < e < 1$, and certify the finite area directly rather than extending error estimates restricted to $e \le 1/10$. Integer square roots and an independent rational Taylor–power-sum calculation both separate the new area from the old actual area. The contribution is a quantitative specialization with reproducible certificates; no optimality or exhaustive priority claim is made.

1 Statement, model, and comparison

All geometry is Euclidean in $\mathbb{R}^3$. Put $H = \sqrt{2}$ and

$$K = \{(x, y, Hs) : 0 \le s \le 1,\ |x| \le 1 - s,\ |y| \le s\}. \tag{1}$$

This is the closed regular tetrahedron with vertices $(\pm 1, 0, 0)$ and $(0, \pm 1, H)$. For a nonzero vector v, a cylinder means $B + \mathbb{R}v$, where $B \subset v^\perp$ is a compact nondegenerate triangle. Let π_E denote orthogonal projection onto a plane E, and define

$$A_{\min}(K) = \min_{\dim E = 2} |\pi_E K|, \qquad R = \frac{\sum_j |B_j|}{A_{\min}(K)}.$$

The sum counts each base separately, including overlaps. There is no restriction on the axes' tilt angles. The denominator is one common minimum projection area, not a different projection area for each cylinder.

Theorem 1. *The tetrahedron K admits a cover by a labeled family of* 20,000 *cylinders of the above kind. With the construction below at $e = 13/125$ and $N =$ 10,000, its normalized total area satisfies*

$$\frac{4999212211665836198151146}{10^{25}} \le R < \frac{4999212211665836198161146}{10^{25}} < \frac{4999213}{10^7}. \tag{2}$$

Every point of K, including its boundary, is covered. For the published $e = 1/10$, $N =$ 10,000 instance of the same family, with actual normalized area R_{old}, one has

$$R < \frac{4999213}{10^7} < \frac{4999215}{10^7} < R_{\text{old}}. \tag{3}$$

Consequently $R <$ 0.499921585997001774.

OpenAI's angular construction proves a finite triangular-cylinder cover below $1/2$, with the same tetrahedron and normalization [1]. Its displayed theorem uses $m = 2\lceil 2/\varepsilon^2 \rceil$, $0 < \varepsilon \le 1/2000$, and

$$R_{\mathrm{OAI}} = \frac{1}{2} - \frac{13}{6000}\varepsilon^2 + E, \qquad |E| \le 2\varepsilon^4.$$

At $\varepsilon = 1/2000$, this is 16,000,000 cylinders. The stated remainder implies

$$R_{\mathrm{OAI}} \ge \frac{1}{2} - \frac{13}{6000 \cdot 2000^2} - \frac{2}{2000^4} > 0.4999215.$$

Thus Theorem 1 improves that explicit realization in both count and actual area. This comparison is with the displayed source instance; it does not bound every possible instance of all four OpenAI constructions [1, 2, 4, 3].

Van de Kamp's quartic overlap estimate and finite secant padding yield the much stronger 20,000-cylinder benchmark [5, 6]. The published outward interval is

$$[0.499921585997001674,\ 0.499921585997001774].$$

We retain that construction and prove a strict comparison with its *actual* finite area. A narrower enclosure of the same number would not establish (3).

Instance	Cylinders	Normalized area and comparison status
OpenAI displayed angular instance, 27 Sep. 2026	16,000,000	$\frac{1}{2} - \frac{13}{6000\cdot 2000^2} \pm \frac{2}{2000^4}$; source theorem with explicit remainder.
van de Kamp, $e = 1/10$, 15 Oct. 2026	20,000	$[0.499921585997001674,\ 0.499921585997001774]$; coverage reproduced and area independently recomputed here.
This note, $e = 13/125$, 17 Oct. 2026	20,000	< 0.4999213; exact enclosure in (2).

Table 1: Same closed tetrahedron, perpendicular triangular bases, and common minimum-projection normalization.

2 Normalization and explicit cylinders

Lemma 2. *The tetrahedron in (1) has* $A_{\min}(K) = H$.

Proof. Its four outward face area vectors are $(0, \pm H, -1)$ and $(\pm H, 0, 1)$. For a unit vector u, the forward faces project onto the shadow with disjoint interiors, by assigning each line parallel to u its forward endpoint. The area of a projected face is the absolute scalar product of its area vector with u. The four area vectors sum to zero, so the shadow area is half the sum of their four absolute products:

$$A(u) = \max(H|u_x|, |u_z|) + \max(H|u_y|, |u_z|).$$

Let $U = \max(|u_x|, |u_z|/H)$ and $V = \max(|u_y|, |u_z|/H)$. Then $U^2 \ge u_x^2$, $V^2 \ge u_y^2$, and $2UV \ge 2u_z^2/H^2 = u_z^2$. Thus $(U+V)^2 \ge 1$, so $A(u) \ge H$. The direction $u = (1, 0, 0)$ gives equality. □

This is the equal-partition case of [5, Sections 1–3]. We reproduce its proof to make the parameter domain explicit. Take any $0 < e < 1$ and integer $N \ge 1$. For $0 \le i < N$, define

$$a = \frac{1}{N}, \quad d = 2a, \quad l_i = -1 + \frac{2i}{N}, \quad r_i = l_i + d, \quad c_i = \frac{l_i + r_i}{2}, \quad I_i = [l_i, r_i], \tag{4}$$

$$G(q) = \frac{1+q^2}{2}, \quad D(q) = \frac{1-q^2}{2}, \quad C(q) = 1 + e^2q^2G(q) + \frac{9}{16}e^4, \tag{5}$$

$$A_i = \frac{1+l_ir_i}{2} = G(c_i) - \frac{a^2}{2}, \qquad M_* = 1 + e^2 + \frac{9}{16}e^4, \tag{6}$$

$$K_0 = M_*\left(e + \frac{e^2}{2}\right) + 2e^2, \qquad \delta = \frac{K_0a + M_*e^2a^2/2}{1-e}, \tag{7}$$

$$M_i = C\big(\max(|l_i|, |r_i|)\big) + \delta. \tag{8}$$

Since C is even and increasing on $[0,1]$, $M_i = \max_{q\in I_i} C(q) + \delta > 0$. Use the intercept triangles

$$P_i = \{(0, qt, Ht) : q \in I_i,\ 0 \le t \le M_i/2\}, \tag{9}$$

$$Q_i = \{(pt, 0, H(1-t)) : p \in I_i,\ 0 \le t \le M_i/2\}, \tag{10}$$

and the axes

$$v_i = (1, eA_i, Hec_i), \qquad w_i = (-eA_i, 1, Hec_i). \tag{11}$$

The cylinders are $P_i + \mathbb{R}v_i$ and $Q_i + \mathbb{R}w_i$, with bases $B_i^- = \pi_{v_i^\perp} P_i$, $B_i^+ = \pi_{w_i^\perp} Q_i$. For example P_i is the convex hull of $(0,0,0)$, $(0, l_iM_i/2, HM_i/2)$, and $(0, r_iM_i/2, HM_i/2)$. It is a compact nondegenerate triangle because $dM_i > 0$. Orthogonal projection is injective on its intercept plane: an axis-parallel vector in $x = 0$ must vanish, as the first coordinate of v_i is one. The analogous argument uses $y = 0$ and w_i. Both projected triangles remain nondegenerate. Projection removes an axis-parallel component, so it preserves the cylinder.

3 Coverage of the entire closed tetrahedron

3.1 Angular selection and exact membership

Fix $z = (x, y, Hs) \in K$. Its axis-parallel intercept in $x = 0$ is

$$z - xv_i = (0, y_0, Ht_0), \qquad y_0 = y - exA_i, \quad t_0 = s - exc_i.$$

Since the first coordinate of v_i is one, this intercept is unique. Thus, including when $t_0 = 0$,

$$z \in P_i + \mathbb{R}v_i \iff 0 \le t_0 \le M_i/2, \quad l_it_0 \le y_0 \le r_it_0. \tag{12}$$

Likewise $z - yw_j = (x_0, 0, H(1-s_0))$, where $s_0 = 1 - s + eyc_j$, $x_0 = x + eyA_j$, so

$$z \in Q_j + \mathbb{R}w_j \iff 0 \le s_0 \le M_j/2, \quad l_js_0 \le x_0 \le r_js_0. \tag{13}$$

Let $q_k = -1 + 2k/N$, $0 \le k \le N$, and $F_k = sq_k + exD(q_k)$. Since $D(-1) = D(1) = 0$, we have $F_0 = -s \le y \le s = F_N$. Choose the first $k \in \{1, \dots, N\}$ with $F_k \ge y$. Then $F_{k-1} \le y$, including when $y = F_0$. Set $i = k - 1$. The identities

$$A_i - l_ic_i = D(l_i), \qquad A_i - r_ic_i = D(r_i)$$

convert $F_{k-1} \le y \le F_k$ to $l_it_0 \le y_0 \le r_it_0$. As $r_i - l_i > 0$, these inequalities imply $t_0 \ge 0$. No monotonicity of F_k is needed.

Apply the same argument to $E_k = (1-s)q_k - eyD(q_k)$. Its endpoints bracket x, since $|x| \le 1 - s$. We obtain a cell j with $s_0 \ge 0$ and $l_js_0 \le x_0 \le r_js_0$. If $t_0 = 0$, then $y_0 = 0$ and (12) gives coverage. If $s_0 = 0$, then $x_0 = 0$ and (13) does so. Assume both are positive before dividing.

3.2 A weighted identity

Set

$$\rho = 2t_0, \quad \sigma = 2s_0, \quad q = y_0/t_0 \in I_i, \quad p = x_0/s_0 \in I_j, \quad X = 2x, \quad Y = 2y.$$

The intercept equations become

$$Y = \rho q + eXA_i, \quad X = \sigma p - eYA_j, \quad \rho + \sigma = 2 + e(Yc_j - Xc_i). \tag{14}$$

Let $\Lambda = 1 + e^2 A_i A_j$. The first two equations give $\Lambda X = \sigma p - eA_j \rho q$ and $\Lambda Y = \rho q + eA_i \sigma p$. Substitution in the third yields

$$w_\rho \rho + w_\sigma \sigma = 2\Lambda, \tag{15}$$

where

$$w_\rho = 1 - eqc_j + e^2 A_j(A_i - qc_i), \qquad w_\sigma = 1 + epc_i + e^2 A_i(A_j - pc_j).$$

The affine function $L_i(q) = A_i - qc_i$ interpolates $D(l_i), D(r_i) \geq 0$, and $0 \leq A_i, A_j \leq 1$. All angular variables and centers have absolute value at most one. Therefore

$$w_\rho, w_\sigma \geq 1 - e > 0. \tag{16}$$

3.3 Continuous overlap and finite padding

For $p, q \in [-1, 1]$, write

$$N_0 = 1 + e^2 G(p)G(q), \quad A_- = 1 - epq + e^2 G(p)D(q), \quad B_+ = 1 + epq + e^2 G(q)D(p).$$

The overlap certificate of [5] is

$$\begin{aligned} &A_- C(q) + B_+ C(p) - 2N_0 \\ &\quad = \frac{e^2}{2}\left(p^2 - q^2 + \frac{epq(1 + p^2 + q^2)}{2}\right)^2 \\ &\qquad + e^4\left[G(p)G(q)\{p^2 D(p) + q^2 D(q)\} + \frac{9 - p^2 q^2 (1 + p^2 + q^2)^2}{8}\right] \\ &\qquad + \frac{9e^6}{16}\{G(p)D(q) + G(q)D(p)\}. \end{aligned} \tag{17}$$

To verify it directly, put $\Delta = p^2 - q^2$, $S = 1 + p^2 + q^2$. Multiplication of the left side gives

$$\begin{aligned} &\frac{e^2 \Delta^2}{2} + \frac{e^3 pq \Delta S}{2} \\ &\quad + e^4\left[\frac{9}{8} + G(p)G(q)\{p^2 D(p) + q^2 D(q)\}\right] + \frac{9e^6}{16}\{G(p)D(q) + G(q)D(p)\}. \end{aligned}$$

Completing the square proves (17). All terms on its right are nonnegative: $G, D \geq 0$, $p^2 q^2 \leq 1$, and $S \leq 3$.

For the secant perturbation, put $b = q - c_i$, so $|b| \leq a$. Then

$$L_i(q) = D(q) - E_i, \quad E_i = \frac{a^2 - b^2}{2}, \quad 0 \leq E_i \leq \frac{a^2}{2}, \quad |A_i - G(q)| \leq |c_i| a + a^2 \leq a.$$

The last inequality uses $|c_i| + a \leq 1$. The corresponding statements hold for $p \in I_j$. Using $D \leq 1/2$ and $G, A_i, A_j \leq 1$, we get

$$|w_\rho - A_-|, \ |w_\sigma - B_+| \leq ea + \frac{e^2}{2}(a + a^2), \tag{18}$$

$$|\Lambda - N_0| \le 2e^2 a. \tag{19}$$

For example the quadratic part of the first bound follows from

$$A_j L_i(q) - G(p)D(q) = (A_j - G(p))D(q) - A_j E_i,$$

and its linear part is at most $e|q||p - c_j| \le ea$. Changing the two factors A_i, A_j one at a time proves (19). Since $0 < C \le M_*$, equations (17)–(19) imply

$$w_\rho C(q) + w_\sigma C(p) - 2\Lambda \ge -2K_0 a - M_* e^2 a^2.$$

But the definition of δ and (16) give

$$(w_\rho + w_\sigma)\delta \ge 2(1-e)\delta = 2K_0 a + M_* e^2 a^2.$$

As $M_i \ge C(q) + \delta$ and $M_j \ge C(p) + \delta$, it follows that

$$w_\rho M_i + w_\sigma M_j \ge 2\Lambda. \tag{20}$$

If both cylinders missed z, their angular membership conditions would already hold, forcing $\rho > M_i$ and $\sigma > M_j$. The positive weights would contradict (15) and (20).

Proposition 3. *For every $0 < e < 1$ and integer $N \ge 1$, these $2N$ cylinders cover all of the closed tetrahedron K.*

All selection and membership inequalities were closed, with zero radii handled before division. Thus the proof includes vertices, faces, edges, grid nodes, and the interface between families. No sampling is used.

4 Area and exact arithmetic certificates

Each intercept triangle has physical area $HdM_i^2/8$. Projection multiplies area by the absolute scalar product of the intercept-plane and projection-plane unit normals, namely $1/\sqrt{Z_i}$, where

$$Z_i = |v_i|^2 = |w_i|^2 = 1 + e^2(A_i^2 + 2c_i^2).$$

The two families contribute equally. Lemma 2 gives

$$R(e, N) = \frac{1}{2N} \sum_{i=0}^{N-1} \frac{M_i^2}{\sqrt{Z_i}}. \tag{21}$$

This exact identity holds for $0 < e < 1$. It does not use the source's later integral approximation, stated only for $e \le 1/10$.

4.1 Integer square-root enclosures

Take $N = 10{,}000$, $e = 13/125$. The padding is exactly

$$\delta = \frac{161414920845251281}{10937500000000000000000}.$$

Let $S = 10^{25}$, $b_i = M_i^2/(2N)$, and express $b_i^2/Z_i = u_i/v_i$ as a positive rational number. Define

$$k_i = \operatorname{isqrt}\left(\left\lfloor S^2 u_i / v_i \right\rfloor\right).$$

Here $\operatorname{isqrt}(n)$ is the greatest nonnegative integer whose square is at most n. The integer inequalities

$$k_i^2 v_i \le S^2 u_i < (k_i + 1)^2 v_i$$

give $k_i/S \le b_i/\sqrt{Z_i} < (k_i + 1)/S$. The implementation checks them for every cell using exact integers. It obtains

e	$\sum_i k_i$	$\sum_i (k_i+1)$
13/125	49992122116658361981511 46	49992122116658361981611 46
1/10	49992158599700167696003 78	49992158599700167696103 78

The first row proves (2); the second proves

$$\frac{4999215859970016769600378}{10^{25}} \le R_{\text{old}} < \frac{4999215859970016769610378}{10^{25}}. \tag{22}$$

Comparison (3) now follows by integer arithmetic. The generator and certificate are `code/retuned_secants.py` and `checkers/retuned_certificate.json`.

4.2 An independent Taylor–power-sum check

A second implementation does not import the constructor, use integer square roots, or read the first certificate. Reflection sends cell i to $N-1-i$, negating c_i and preserving A_i, M_i, Z_i. Since $N = 10{,}000$ is even and zero is a node, exactly half the cells are positive. For $1 \le k \le 5000$, put

$$q = \frac{k}{5000}, \quad c = \frac{2k-1}{10000}, \quad A = \frac{1}{2}\left(1 + \frac{k(k-1)}{25000000}\right),$$
$$M = 1 + \frac{e^2}{2}(q^2+q^4) + \frac{9}{16}e^4 + \delta, \qquad z = e^2(A^2+2c^2).$$

Thus $R(e,10000) = \frac{1}{10000}\sum_{k=1}^{5000} M^2(1+z)^{-1/2}$. Define

$$T_4(z) = 1 - \frac{z}{2} + \frac{3z^2}{8} - \frac{5z^3}{16} + \frac{35z^4}{128}, \qquad T_5(z) = T_4(z) - \frac{63z^5}{256}.$$

The j-th derivative of $(1+z)^{-1/2}$ has sign $(-1)^j$ on $[0,\infty)$. Taylor's theorem with remainder gives $T_5(z) \le (1+z)^{-1/2} \le T_4(z)$ for $z \ge 0$. The weighted sums are polynomials in k with rational coefficients.

For $S_j(m) = \sum_{k=1}^{m} k^j$, telescoping $(k+1)^{j+1} - k^{j+1}$ gives

$$S_0(m) = m, \qquad S_j(m) = \frac{(m+1)^{j+1} - 1 - \sum_{h=0}^{j-1}\binom{j+1}{h} S_h(m)}{j+1}.$$

Exact polynomial multiplication and this recurrence produce

$$0.4999212211562773 87 \le R(13/125, 10000) \le 0.4999212215782042 04,$$
$$0.4999215859905606 69 \le R(1/10, 10000) \le 0.4999215862752333 98.$$

These are outward-rounded decimal intervals; they independently prove (3). The unrounded fractions are in `checkers/power_sum_bounds.json`; the implementation is `checkers/check_retuned.py`. Either certificate, together with Proposition 3 and the nondegeneracy argument, proves Theorem 1.

5 Reproducibility and proof dependencies

The package uses Python 3.8 or later and only its standard library; the recorded runs used Python 3.12.3. No random seed, floating-point arithmetic, network connection, or stored cylinder list is needed. From the directory containing this paper, run:

```
python3 code/reproduce.py
python3 code/retuned_secants.py --index 0
python3 code/retuned_secants.py --index 19999
latexmk -pdf -jobname=paper -interaction=nonstopmode main.tex
```

The first command recomputes both certificates, checks the archived outputs, and checks the overlap identity with the separate reviewer's exact polynomial implementation. It also checks the rational comparison with the displayed OpenAI instance. Failure gives a nonzero exit code. These computations complement the universal geometric proof.

The indexed generator returns rational data in coordinates $(x, y, z/H)$, whose physical metric is $\mathrm{diag}(1, 1, 2)$. A perpendicular vertex is computed by $P \mapsto P - \langle P, v\rangle v/\langle v, v\rangle$. Indices $0, \ldots, 9999$ designate the lower family; $10000, \ldots, 19999$ designate the upper family. These formulas specify all triangles without a large coordinate file.

6 Sources and construction

The minimum-projection formula, compatible secant axes, and shared-side first crossing come from OpenAI's angular construction. The quartic cap, overlap identity, and finite padding come from van de Kamp [1, 5]. This paper chooses $e = 13/125$ and certifies its finite area by exact rational arithmetic. Coverage is proved on $0 < e < 1$, including the parameter used here. The corresponding axis tilt varies with e.

References

[1] OpenAI. Finite angular cylinder covers below the half-area bound. OpenAI Math Release preprint OAI:Finite-angular-cylinder-covers-below-the-half-area-bound-September-27-2026, 2026.

[2] OpenAI. Finite cylinder approximation of ruled sets. OpenAI Math Release preprint OAI:Finite-cylinder-approximation-of-ruled-sets-September-27-2026, 2026.

[3] OpenAI. Finite triangular approximation of radial sweeps. OpenAI Math Release preprint OAI:Finite-triangular-approximation-of-radial-sweeps-September-27-2026, 2026.

[4] OpenAI. Slope-field perturbations of the two-cylinder covering. OpenAI Math Release preprint OAI:Slope-field-perturbations-of-the-two-cylinder-covering-September-27-2026, 2026.

[5] shjvandekamp. A quartic overlap theorem with effective finite secant covers. Repository manuscript, SECANT-THEOREM.md, 2026. Revision db6e1d89a0637b013a86464bd766c0a5120e454e; inspected 17 October 2026. https://github.com/shjvandekamp/math/blob/db6e1d89a0637b013a86464bd766c0a5120e454e/topics/cylinder-covering/SECANT-THEOREM.md.

[6] shjvandekamp. Result: an effective quartic/secant cover of the regular tetrahedron. Repository manuscript, RESULT.md, 2026. Revision 8b63303107ca07dce2e72ed65b11d3af75540e5f; inspected 17 October 2026. https://github.com/shjvandekamp/math/blob/8b63303107ca07dce2e72ed65b11d3af75540e5f/topics/cylinder-covering/RESULT.md.

Mathematical supplement

The proof supplements, exact certificate data and arithmetic programs referenced by filename are included in the attached mathematical supplement (ZIP). Open the attachment using a PDF reader with file-attachment support.